



SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

Addis Ababa University
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PLACENTA PREVIA

Dr. Hussen T (MD, OBGYN Resident)

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OUTLINE

- Introduction
- Epidemiology
- Risk factors
- Pathogenesis
- Diagnosis
- Management
- Summary

INTRODUCTION

- **Placenta previa** refers to the presence of placental tissue that extends over the internal cervical os(with in 2cm).
- Placenta previa should be suspected in any pregnant woman beyond 20 weeks of gestation who presents with vaginal bleeding.
 - US determination of the placental location **before** a digital vaginal examination is performed

EPIDEMIOLOGY

- The prevalence of placenta previa is **4-5**/1000 births but varies worldwide.
- The prevalence is much higher at 20 weeks of gestation than at delivery.

RISK FACTORS

- Previous placenta previa: **4-8%** recurrence rate
- Previous cesarean birth: **47-60%** ↑ in the risk of PP in a subsequent px
- Multiple gestation: prevalence of PP was 40% higher among twin births than singleton
- Other risk factors:
 - AMA
 - Multiparity
 - Previous uterine surgical procedure

CONT'D

- The risk of previa implantation ↑ with the number of **prior CDs**.
- The **rate of placenta previa** at 2nd births for women with:
 - Vaginal 1st births: **4.4**/1000 births
 - Cesarean 1st birth: **8.7**/1000 births

PATHOGENESIS

Two key factors in the pathogenesis of pp are:

① Decidual factor:

- Suboptimally vascularized decidua.
 - Promote implantation of trophoblast

② Placental factor:

- Large placental surface area
 - ↑chances that the placenta will cover the cervical os

PLACENTAL MIGRATION(**MISNOMER**)

- **90%** of PP identified on US at **18 - 20wks** of gestation resolve before delivery at term.
- **Two theories** have been proposed to account for this phenomenon:

① Differential growth (Tropho-tropism)

- The placenta preferentially grows towards more vascular fundus.
- This leads to progressive **unidirectional growth** of trophoblastic tissue toward the fundus.

② Formation/Lengthening of the LUS

- From **5mm@20wks**
- To **>50mm @term**
- This dev't of the LUS **relocates** the lower edge of the placenta away from the internal os.

CONT'D

- The possibility of persistence is determined by :
 1. **GA at diagnosis**
 - **90%** of PP identified on US at **18 - 20wks** resolve before delivery at term.
 2. **Degree of extension over the os**
 3. **Anterior vs posterior position:**
 - Anterior PP is **more likely to resolve** with advancing gestation than posterior PP.
 4. **Placental thickness:**
 - Thick midtrimester placenta that covers the internal os is more likely to persist than a thin midtrimester placenta previa.

① GA AT DIAGNOSIS

TABLE 18.2 Potential for Placenta Previa at Term by Gestational Age at Diagnosis

Gestational Age at Diagnosis (Weeks)	Previa at Term (%)
15–19	12
20–23	34
24–27	49
28–31	62
32–35	73

② DEGREE OF EXTENSION OVER THE OS

- The more the placenta extends over the os, the more likely it is to persist until delivery.
- At **18 - 24 wks** of gestation, pooled data suggest that:

Extension over internal os	Probability of PP at delivery
<15mm	0%
15-25mm	20%
25-55mm	40-100%
>55mm	100%

MECHANISM OF BLEEDING

Gradual changes in the LUS & Cx

Uterine contraction

Shearing forces to the **inelastic** placental attachment site

**Partial detachment &
Placental bleeding**

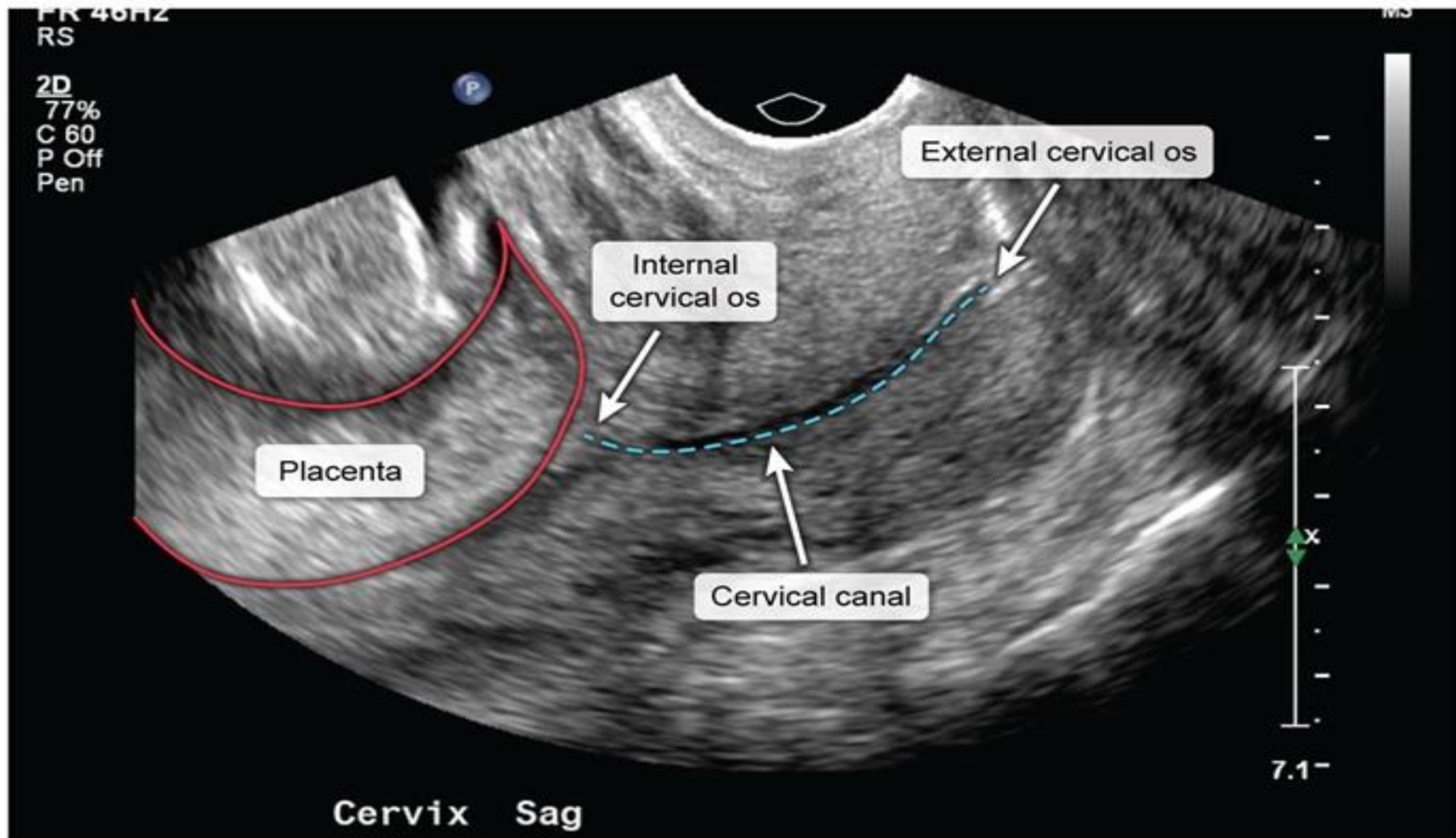
CLINICAL PRESENTATION

- Among patients with persistent placenta previa:
 - 1/3rd have an initial bleeding episode before 30 wks
 - 1/3rd become symptomatic between 30-36 wks.
 - 1/3rd have their first bleeding after 36 wks.
 - ~10 % reach term without bleeding

DIAGNOSIS

- Clinical
- Ultrasonography
 - TAS
 - TVS

Complete placenta previa



Other associated findings

- Malpresentation
- Vasa previa and velamentous umbilical cord
- Fetal growth restriction
- Congenital anomalies

MANAGEMENT

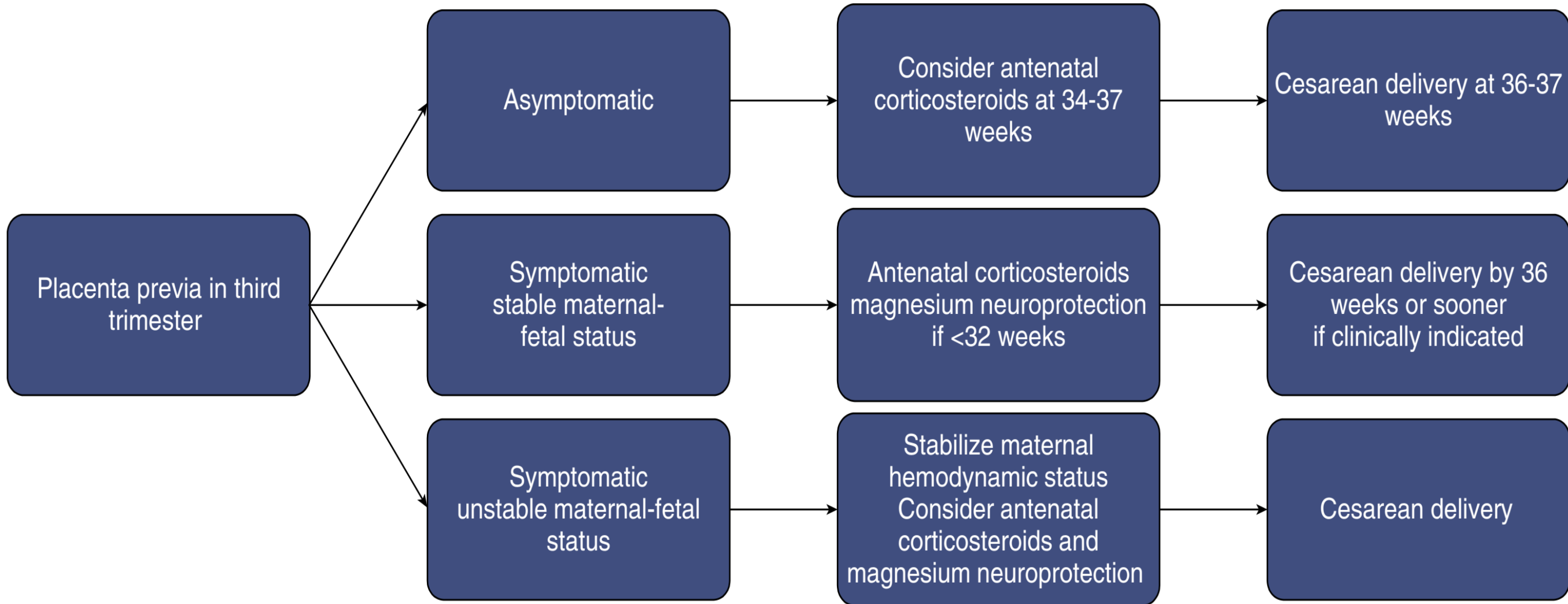


Fig. 18.6 Algorithm for Management of Placenta Previa.

RECURRENCE

- PP recurs in **4-8%** of subsequent pregnancies

Risk factors	• Prior placenta previa • Prior cesarean delivery • Multiple gestation
Clinical features	• Painless vaginal bleeding >20 weeks gestation
Diagnosis	• Transabdominal followed by transvaginal sonogram
Management	• No intercourse • No digital cervical examination • Inpatient admission for bleeding episodes

REFERENCES

- Williams Obstetrics 26th edition
- Gabbe's Obstetrics 9th edition

THANK YOU